

Research Proposal

On

Implementation of multi - model coding tutor and collaboration platform by using Generative AI.

A Research Proposal submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

In

Computer Science & Engineering

Submitted by

Arya Thool

Kajal Khandekar

Arya Surkar

Abhishek Gupta

Under the Guidance of

Prof. Anuja Ghasad

Assistant Professor



Education to Eternity

Department of Computer Science & Engineering

J D College of Engineering and Management, Nagpur-441501

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai.

Year 2025-26

CERTIFICATE

This is to certify that the Research Proposal entitled, **“Implementation of multi - model coding tutor and collaboration platform by using Generative AI.”** in **Computer Science & Engineering** in the faculty of Science and Technology submitted by **Arya Thool, Kajal Khandekar, Arya Surkar, Abhishek Gupta** to **Dr. Babasaheb Ambedkar Technological University, Lonere** for the award of the degree of **Bachelor of Technology** is a bonafide record of work carried out by them under my supervision. The contents of this Research proposal, in full or in parts, have not been submitted or published to any other Institute or University for the award of any degree or diploma.

**Project Guide, CSE
JDCOEM, Nagpur**

Forwarded to:

**Project Coordinator
CSE, JDCOEM, Nagpur**

**HOD CSE
JDCOEM, Nagpur**

CERTIFICATE OF APPROVAL

This is to certify that the Research Proposal entitled on **Implementation of multi - model coding tutor and collaboration platform by using Generative AI** is approved work done by

Arya Thool

Kajal Khandekar

Arya Surkar

Abhishek Gupta

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering** at J D College of Engineering & Management, Nagpur affiliated to **Dr. Babasaheb Ambedkar Technological University, Lonere** during the academic year 2025-26.

**Guide, CSE
JDCOEM, Nagpur**

**HOD CSE
JDCOEM, Nagpur**

Project Examination held on _____

Internal Examiner

External Examiner

**Project Coordinator
CSE, JDCOEM, Nagpur**

Index

Sr. No.	Content	Page No.
	Abstract	1
1.	INTRODUCTION	2
2.	LITERATURE SURVEY	3
3.	RESEARCH GAP	7
4.	PROBLEM STATEMENT	8
5.	AIM & OBJECTIVES	9
6.	SCOPE	10
8.	LIMITATION	11
9.	METHODOLOGY	12
10.	FLOW CHART	13
11.	CONCLUSION	14
	REFERENCES	15
	Plagiarism Report	17
	Grammar Report	18

Research Proposal

Implementation of multi - model coding tutor and collaboration platform by using Generative AI.

Abstract

Learning programming effectively requires more than syntax memorization—it demands logical thinking, conceptual clarity, and the ability to translate ideas into functional code. However, conventional teaching platforms often rely on single-mode content delivery, which limits interaction, personalization, and collaboration. The proposed project, *Implementation of Multi-Modal Coding Tutor and Collaboration Platform by Using Generative AI*, aims to address these challenges by developing an intelligent, interactive learning environment that combines multiple input modes such as text, flowcharts, and voice. The system utilizes Generative AI models to interpret learner intent, synthesize code, and provide real-time, step-by-step explanations and contextual feedback. Additionally, it integrates a collaborative coding workspace that supports peer-to-peer interaction and mentor guidance in real time. By unifying multimodal learning, explainable AI, and collaborative problem-solving, the platform seeks to enhance programming comprehension, reduce debugging effort, and foster an engaging and scalable learning experience for students.

1. INTRODUCTION

Programming is a fundamental component of computer science education, enabling learners to apply computational thinking to solve real-world problems. However, many students find it difficult to grasp the logical structure behind programming and often focus mainly on syntax rather than reasoning. Traditional learning methods—such as textbooks, video tutorials, and static coding exercises—present concepts in isolation, offering limited interaction and feedback. This leads to fragmented understanding and slower learning progress.

A common challenge faced by students is converting their ideas, algorithms, or flowcharts into executable code. The absence of instant, contextual feedback and opportunities for collaboration further restricts their ability to learn effectively. As a result, learners struggle to connect theoretical concepts with practical implementation.

To address these issues, the proposed project titled **“Implementation of Multi-Modal Coding Tutor and Collaboration Platform by Using Generative AI”** aims to create an intelligent, interactive, and collaborative environment for programming education. The system integrates multiple input modes such as text, diagrams, and voice to support diverse learning preferences. A Generative AI-based logic engine interprets learner intent, generates relevant code, and provides step-by-step explanations to enhance comprehension. Additionally, a real-time collaborative coding workspace allows students and mentors to interact, share ideas, and solve problems together.

By combining **multi-modal learning, explainable AI, and real-time collaboration, LogicFlow AI** this project seeks to bridge the gap between theoretical knowledge and practical coding skills, ultimately improving engagement, understanding, and teamwork in programming education.

2. LITERATURE SURVEY

[1] Sharma, P., & Harkishan, M. (2022). *Designing an Intelligent Tutoring System for Computer Programming in the Pacific. Education and Information Technologies.*

This study proposes an intelligent tutoring system that leverages optimized algorithms to provide personalized learning support for programming students. It focuses on adapting to individual learning needs, enhancing programming skill acquisition, and providing interactive feedback to accelerate learning outcomes.

[2] Vali, S. N., Vikas, M., Smaran, V., Augusteen, E., & Varun, N. (2024). *AI Powered Code Reviewer. International Journal of Research Publication and Reviews.*

This paper presents a web-based AI code reviewer built on the MERN stack. The system analyzes submitted code to identify syntax errors, suggest improvements, and offer targeted feedback, effectively assisting developers and learners in improving code quality and programming efficiency.

[3] Sengul, C., Neykova, R., & Destefanis, G. (2024). *Software Engineering Education in the Era of Conversational AI. PLOS Digital Health.*

The paper explores the integration of conversational AI tools in software engineering education. It discusses their role in teaching requirements, design, and testing, emphasizing how AI agents can support both instructors and learners by providing interactive explanations and practical assistance during learning activities.

[4] Rasheed, Z., et al. (2024). *AI-Powered Code Review with LLMs: Early Results. arXiv preprint.*

This work introduces a multi-agent large language model system designed for automated code review. It demonstrates improved bug detection, code optimization suggestions, and contextual feedback compared to conventional static analysis tools, highlighting the potential of LLMs for real-time developer support.

[5] Dhannur, S., T. M. D., Gowda, Y., B. K. Y., & Chaithra, P. (2025). *Collaborative Coding Platform. International Journal of Creative Research Thoughts.*

The study presents a collaborative platform that enables multiple users to code simultaneously in real time. It focuses on peer-to-peer interaction, live code

synchronization, and integrated communication tools to improve remote teamwork and enhance collaborative problem-solving in programming environments.

[6] Kovari, A., et al. (2025). *A Systematic Review of AI-Powered Collaborative Learning in Higher Education. Computers & Education: Artificial Intelligence.* This systematic review examines how AI enhances collaborative learning in higher education. It categorizes approaches to AI integration, evaluates their effectiveness in promoting knowledge exchange, and identifies strategies that improve engagement, group work, and peer learning dynamics in academic settings.

[7] Raihan, N., Siddiq, M. L., Santos, J. C. S., & Zampieri, M. (2025). *Large Language Models in Computer Science Education: A Systematic Literature Review. VL/HCC Conference Proceedings.*

This review provides a comprehensive overview of how large language models are being adopted in computer science education. It discusses their role in personalized tutoring, enhancing student engagement, automating assessments, and providing intelligent support, while also outlining challenges related to reliability and pedagogy.

[8] Ogunleye, B., Zakariyyah, K. I., Ajao, O., Olayinka, O., & Sharma, H. (2025). *A Systematic Review of Generative AI for Teaching and Learning Practice. Education Sciences.*

This paper synthesizes findings from a wide range of studies examining the role of generative AI in educational practice. It explores its potential in adaptive teaching, content creation, student engagement, and skill development, while also addressing implementation challenges and ethical considerations.

[9] Wang, T., et al. (2023). *Exploring the Role of AI Assistants in Computer Science Education. IEEE Xplore.*

The study investigates how AI assistants, such as ChatGPT, can support computer science education by solving programming problems, assisting in classroom activities, and enhancing the teaching-learning process. It also examines instructor perspectives on integrating AI tools into curricula effectively.

[10] Maity, S., & Deroy, A. (2024). *Generative AI and Its Impact on Personalized Intelligent Tutoring Systems*. *arXiv preprint*.

This paper explores how generative AI can transform intelligent tutoring systems by enabling personalized learning experiences. It emphasizes adaptive learning paths, real-time feedback, and the creation of dynamic tutoring environments that adjust to individual learner needs.

[11] Calò, T., & MacLellan, C. J. (2024). *Towards Educator-Driven Tutor Authoring: Generative AI Approaches for Creating Intelligent Tutor Interfaces*. *arXiv preprint*.

The research focuses on empowering educators to create adaptive tutor interfaces using generative AI. It proposes methods that reduce the technical barriers for instructors, allowing them to design interactive, AI-driven learning tools that adapt to students' progress and learning styles.

[12] Lin, K. Y., Cheng, P. H., & Chen, L. W. (2025). *AI-Powered Pedagogical Frameworks: Advancing Computational Thinking and Coding Proficiency in Secondary Education*. *Research Square Preprint*.

This work proposes an AI-driven pedagogical framework that uses inquiry-based learning and Socratic questioning techniques to enhance students' computational thinking and coding skills. It focuses on how AI can support structured reasoning and deeper conceptual understanding in secondary education.

[13] Gaikwad, A., Gupta, A., Yeole, P., Barde, S., Bhosale, N., & Baste, P. (2024). *A Collaborative Code Platform with Advanced AI Features and Real-Time Collaboration Tools*. *International Journal for Research in Applied Science & Engineering Technology*.

This study develops a collaborative coding environment equipped with AI-driven analysis, multilingual support, and real-time synchronization. It aims to facilitate dynamic teamwork among learners, bridging geographic and linguistic barriers in collaborative software development.

[14] Lan, Y., Li, X., Du, H., Lu, X., Gao, M., Qian, W., & Zhou, A. (2024). *Survey of Natural Language Processing for Education: Taxonomy, Systematic Review, and Future Trends*. *arXiv preprint*.

The paper presents a structured taxonomy of NLP applications in education, covering areas such as automated feedback, assessment systems, and language model integration. It highlights emerging trends and research directions for applying NLP techniques in intelligent tutoring and learning analytics

[15] Zerkouk, M., Mihoubi, M., & Chikhaoui, B. (2024). *A Comprehensive Review of AI-Based Intelligent Tutoring Systems: Applications and Challenges*. *arXiv preprint*.

This review provides a detailed mapping of recent developments in AI-based intelligent tutoring systems. It examines pedagogical models, NLP methods, and adaptive learning mechanisms, while identifying challenges related to scalability, explainability, and evaluation.

3. RESEARCH GAP

1. **Lack of Integrated Multimodal Learning Environments:**

Most existing intelligent tutoring systems rely heavily on text-based or code-centric interactions. There is a lack of research into unified multimodal learning frameworks that combine **text, diagrams, and voice-based inputs** to support diverse learning preferences and cognitive styles.

2. **Limited Real-Time, Context-Aware AI Feedback:**

Current coding tutors offer generic or post-task suggestions rather than **real-time, adaptive guidance**. There is insufficient exploration of AI systems capable of analyzing learners' actions dynamically and generating **contextual, step-by-step feedback** during live coding sessions.

3. **Insufficient Collaboration-Centric AI Solutions:**

Although several studies highlight the importance of peer-assisted learning, few focus on **real-time AI-supported collaboration** that synchronizes code, logic visualization, and shared feedback in a multi-user environment.

4. **Explainability and Transparency Deficiency:**

Many AI-driven tutors function as “black boxes,” offering results without clear reasoning. There is a research gap in **explainable AI (XAI)** mechanisms that can make model decisions interpretable to learners and educators, fostering trust and deeper comprehension.

5. **Ethical and Fairness Considerations in Educational AI:**

Limited work has been done on addressing **bias, fairness, and data privacy** issues in collaborative learning environments powered by Generative AI. Ensuring ethical use of learner data and equitable treatment across skill levels remains an underexplored area.

4. PROBLEM STATEMENT

Programming education requires more than learning syntax; it demands logical thinking and the ability to convert abstract ideas into functional code. However, most current learning platforms focus on syntax drills or isolated exercises, offering limited real-time feedback and minimal collaboration. Students often struggle to connect theoretical logic with practical implementation, slowing progress and reducing engagement. Furthermore, existing tools rarely incorporate multimodal learning methods that accommodate diverse learning preferences such as visual, textual, and auditory understanding. These challenges highlight the need for an intelligent, interactive, and collaborative AI-driven system like *LogicFlow AI* to make programming education more adaptive, engaging, and effective.

5. OBJECTIVES

- To design and implement a **multi-modal learning platform** that integrates text, voice, and visual diagram inputs to enhance programming logic comprehension.
- To develop a **Generative AI-based semantic logic engine** capable of interpreting learner intent, synthesizing code, and providing step-by-step logical explanations.
- To enable **instant, context-aware feedback** that assists learners in debugging, conceptual understanding, and improving problem-solving skills.
- To create a **real-time collaborative coding environment** that supports peer learning, mentorship, and interactive code sharing.
- To evaluate the effectiveness of **LogicFlow AI** through usability testing, learner feedback, and comparative analysis with traditional coding platforms.

6. SCOPE

The proposed project focuses on developing an intelligent, multi-modal coding tutor and collaboration platform, **LogicFlow AI**, that enhances programming logic learning through the integration of text, voice, and diagram-based inputs. The system's scope includes the implementation of a **semantic logic understanding engine** capable of interpreting user intent, translating conceptual designs into executable code, and providing step-by-step logical explanations. It will also incorporate a **real-time collaborative editor** that supports pair programming, peer-to-peer interaction, and instructor guidance within shared sessions. The project further encompasses **usability testing**, **performance benchmarking**, and **comparative evaluation** against conventional learning platforms to validate its educational effectiveness. However, the scope is limited to **teaching programming logic** and **collaborative learning**, rather than complete automated code generation or software development functionalities.

7. LIMITATIONS

- **Complexity of Semantic Understanding:** Translating natural language, voice commands, or visual diagrams into accurate and executable code logic poses significant challenges in semantic interpretation and intent recognition.
- **Integration of Multi-Modal Inputs:** Achieving seamless interaction between text, voice, and diagram-based inputs may lead to integration and consistency issues during real-time operation.
- **Computational Resource Dependency:** The use of Generative AI models for code synthesis and contextual feedback requires substantial computational power, which may limit accessibility on low-end or resource-constrained systems.
- **Real-Time Collaboration Constraints:** Maintaining synchronization, minimizing latency, and handling version conflicts in live collaborative sessions can be technically demanding and require robust system architecture.
- **Evaluation and Dataset Limitations:** The availability of high-quality, diverse datasets for training and testing the AI components may restrict the platform's adaptability across different programming domains.

8. METHODOLOGY

a. Exploratory Research and Requirement Analysis

A detailed literature survey will be conducted on AI-based tutoring systems, program synthesis, and collaborative learning tools to identify functional requirements, design needs, and suitable technologies for LogicFlow AI.

b. System Design and Architecture Development

The architecture will be structured around key modules—multimodal input processing, semantic logic understanding, generative AI integration, and real-time collaboration. The design will emphasize modularity, scalability, and interpretability.

c. Agile Prototyping and Implementation

An iterative development approach will be followed, beginning with core features like AST parsing and flowchart-to-code translation. Subsequent stages will add AI-based logic interpretation, feedback mechanisms, and collaboration tools using frameworks such as Python, Node.js, TensorFlow, and WebRTC.

d. Human-Centric Design and Explainability

Explainable AI (XAI) principles will guide the interface and logic visualization. The system will focus on step-by-step reasoning and conceptual clarity rather than automated code generation.

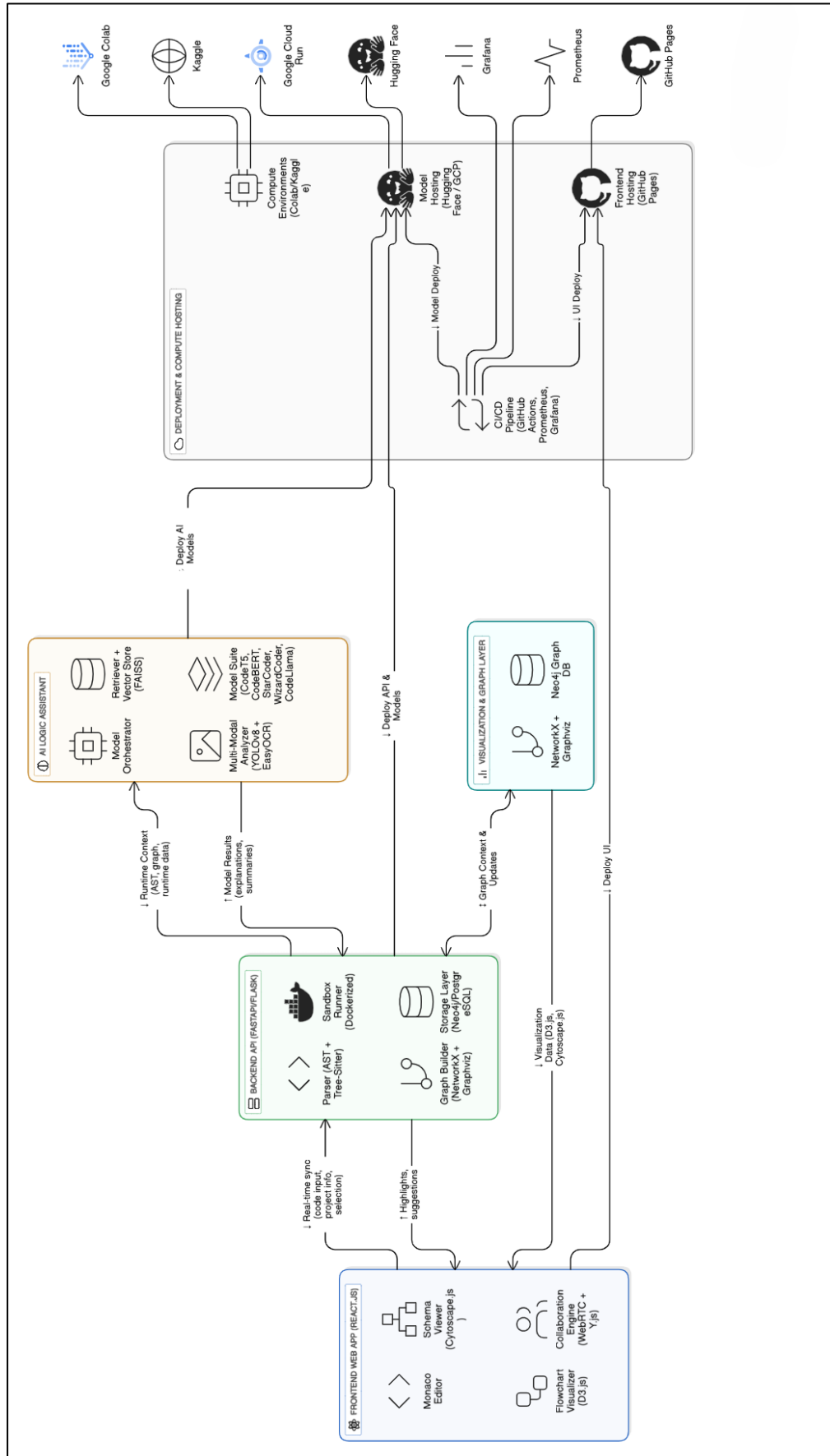
e. Evaluation and Validation

The prototype will undergo user testing, peer evaluation, and comparative benchmarking with existing AI coding tools (e.g., GitHub Copilot, Replit Ghostwriter) to assess usability and learning impact.

f. Deployment and Dissemination

The final version will be deployed through open platforms such as GitHub or Hugging Face, with documentation and results prepared for academic dissemination or potential publication.

9. FLOW CHART



10. CONCLUSION

The proposed project, LogicFlow AI, aims to bridge the gap between learning programming concepts and applying them effectively through code. By integrating multi-modal learning, Generative AI, and real-time collaboration, the platform promotes an interactive and adaptive learning experience. Focusing on explainable and human-centric AI, it emphasizes logical understanding and problem-solving rather than mere code generation. The project is expected to demonstrate the feasibility of combining multimodal education with collaborative learning tools, ultimately contributing to more effective, inclusive, and engaging programming education.

REFERENCE

- [1] P. Sharma and M. Harkishan, “Designing an Intelligent Tutoring System for Computer Programming in the Pacific,” in *Education and Information Technologies*, vol. 27, pages 6197–6209, 2022, doi: 10.1007/s10639-021-10882-9.
- [2] S. N. Vali, M. Vikas, V. Smaran, E. Augusteen, and N. Varun, “AI Powered Code Reviewer,” *International Journal of Research Publication and Reviews*, vol. 5, pp. 100–106, 2024, doi :10.55248/gengpi.6.0825.2930.
- [3] C. Sengul, R. Neykova, G. Destefanis, “Software Engineering Education in the Era of Conversational AI,” *PLOS Digital Health*, vol. 7, pp. 45–55, 2024, doi: 10.3389/frai.2024.1436350.
- [4] Z. Rasheed, et al., “AI-Powered Code Review with LLMs: Early Results,” *arXiv preprint*, vol. 1, 2024, doi: 10.48550/arXiv.2404.18496.
- [5] S. Dhannur, D. T. M., Y. Gowda, Y. B. K., and P. Chaithra, “Collaborative Coding Platform” *International Journal of Creative Research Thoughts*, vol. 13, pp. c711-c719, 2025, doi: _.
- [6] A. Kovari, et al., “A Systematic Review of AI-Powered Collaborative Learning in Higher Education,” *Computers & Education: Artificial Intelligence*, vol. 11, 2025, doi: 10.1016/j.ssaho.2025.101335.
- [7] N. Raihan, M. L. Siddiq, J. C. S. Santos, and M. Zampieri, “Large Language Models in Computer Science Education: A Systematic Literature Review,” *VL/HCC Conference Proceedings*, pp. Pages 938 - 944, 2025, doi: 10.1145/3641554.3701863.
- [8] B. Ogunleye et al., “A Systematic Review of Generative AI for Teaching and Learning Practice,” *arXiv preprint*, vol. 16, 2025, 10.3390/educsci14060636.
- [9] T. Wang, et al., “Exploring the Role of AI Assistants in Computer Science Education,” in *IEEE Xplore*, vol. 5, 2023, doi: 10.1109/VL-HCC57772.2023.00018.

- [10] S. Maity and A. Deroy, “Generative AI and Its Impact on Personalized Intelligent Tutoring Systems,” arXiv preprint, vol. 1, 2024, doi: 10.48550/arXiv.2410.10650.
- [11] T. Calò and C. J. MacLellan, “Towards Educator-Driven Tutor Authoring: Generative AI Approaches for Creating Intelligent Tutor Interfaces,” arXiv preprint, vol. 3, pp. 305 – 309, 2024, doi: 10.1145/3657604.3664694.
- [12] K. Y. Lin, P. H. Cheng, and L. W. Chen, “AI-Powered Pedagogical Frameworks: Advancing Computational Thinking and Coding Proficiency in Secondary Education,” Research Square Preprint, vol. 2, 2025, 10.36227/techrxiv.174910010.00866598/v1.
- [13] A. Gaikwad, A. Gupta, P. Yeole, S. Barde, N. Bhosale, and P. Baste, “A Collaborative Code Platform with Advanced AI Features and Real-Time Collaboration Tools,” International Journal for Research in Applied Science & Engineering Technology, vol. 12, pp. 134–142, 2024, doi: _.
- [14] Y. Lan, X. Li, H. Du, X. Lu, M. Gao, W. Qian, and A. Zhou, “Survey of Natural Language Processing for Education: Taxonomy, Systematic Review, and Future Trends,” arXiv preprint, vol. 3, 2024, doi: 10.48550/arXiv.2401.07518.
- [15] M. Zerkouk, M. Mihoubi, and B. Chikhaoui, “A Comprehensive Review of AI-Based Intelligent Tutoring Systems: Applications and Challenges,” arXiv preprint, vol. 1, 2024, doi: 10.48550/arXiv.2507.18882.

Plagiarism Report

Tool Used: <https://papersowl.com/free-plagiarism-checker>

Document Checked: Research Proposal – *Generative AI-Based Coding Tutor and Collaboration Platform*

Date of Check: 09/10/2025

Total Similarity Index: 5.7%

Originality Score: 94.3%

Matched Sources: 1

Remarks: The report indicates a high level of originality. Minor matches are from general academic phrases and do not indicate plagiarism.

abstract learning programming effectively requires more than syntax memorizationit demands logical thinking conceptual clarity and the ability to translate ideas into functional code however conventional teaching platforms often rely on single-mode content delivery which limits interaction personalization and collaboration the proposed project implementation of multi-modal coding tutor and collaboration platform by using generative ai aims to address these challenges by developing an intelligent interactive learning environment that combines multiple input modes such as text flowcharts and voice the system utilizes generative ai models to interpret learner intent synthesize code and provide real-time step-by-step explanations and contextual feedback

SIMILAR 5.7% **ORIGINAL** 94.3%

MAKE IT UNIQUE

Text matches these sources

Sources: Similarity:

1. <https://www.allassignmenthelp.com...> 5.7%

[Exclude source](#) [View source](#)

☐ I'm not a robot
reCAPTCHA is changing its terms of service

Grammar Report

Tool Used: <https://quillbot.com/grammar-check>

Document Checked: *Research Proposal – Implementation of Multi-Modal Coding Tutor and Collaboration Platform Using Generative AI*

Date Checked: 09 October 2025

Grammar Score: 100 / 100

Overall Writing Score: 85 / 100 (Good)

Remarks: The document demonstrates excellent grammatical accuracy, with strong engagement and fluency. Minor improvements may be possible in clarity and flow, but the overall language quality is well within acceptable academic standards.

